

(a) Repeated-token input

Raw text → tokenizer
prefix + repeated tokens

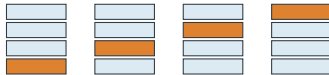


$V = \mathbf{1}v^T$
identical Value rows

Row shuffling leaves
repeated Value rows unchanged.

(b) Observable marker orbit

$$C_j = uw^T + s_j a^T$$



Marker positions before row mixing

$b!$ permutations → b marker states

(c) Transfer within a key epoch

State differences
estimate equivalent transforms

Target cache → row demixing

Public dictionary → token multiset